SMART Service Audit — Customer Value & Business Case

Nov 6, 2026 · @Minh Le

Summary

SMART Service Audit gives a waste-producing customer an independent, explainable check of the waste services it buys, and the weight at source makes it possible. In the synthetic dataset, the value a customer can realistically capture is about 7 % of its waste spend, a median of €38 per device per month, concentrated in large multi-site customers.

Recommended model for Europress:

- **Software tier in SMART**, priced per device per month, sold first to customers with 16+ devices (realisable value ≈ €79 per device per month in the data).
- **Weighing standard on new machines**, where the marginal hardware cost is ≈ €22–49 per device per month over five years; **retrofit only where value is high** (top-quartile devices, ≥ €93 per device per month).
- **Outcome-based pilot option** (25 % of the savings the pilot verifies against invoices and weighbridge data) to lower the first-year barrier, then conversion to the subscription.

Every figure here comes from the Since AI synthetic dataset and our own stated parameters. It shows how value arises and how sensitive it is, not real savings or a Europress price.

Customer problem and value

Today the waste operator weighs, grades and invoices, and the customer has little independent evidence; device weight plus the customer's own contracts and invoices change that. Across the 52 fictional customers (447 devices, €5.19 M annual waste spend), the calculator identifies €1.18 M of value a year, of which we assume €0.37 M is realisable.

Value lever	What the customer gets	Identified (€/yr)	Realisation we assume	Why that share
Operator performance	contest emergency, side-waste and wasted-trip charges with device evidence	406,129	30 %	needs a scorecard the operator accepts
Contract review	material credit on credit-eligible tonnes where the contract gives none	241,692	30 %	needs the operator to agree
Over-servicing	fewer lifts of half-empty containers	231,625	30 %	hygiene, access and peaks limit it
Operator tendering	renegotiate or tender clearly expensive contracts	145,670	40 %	published success odds, notice periods
Contamination	keep material credits by fixing sorting on site	114,002	20 %	needs behaviour change
Billing review	dispute documented invoice discrepancies	44,339	60 %	document-backed; 90 % recovery is published, discounted for errors

The audit engine behind these levers was scored with the official harness: operator scorecard 96.3/100 (baseline 67.8), action regret €12 per contract (baseline €240), anomaly macro-F1 0.82 (baseline 0.78). The scorecard and the actions are computed only from data available at each report or decision date (as-of snapshots, verified by a mutation test).

Who benefits

Value per device is highest for large fleets, and only the top quarter of devices can carry a retrofit kit plus a subscription; factory-fitted weighing is affordable much more widely.

Segment (devices per customer)	Customers	Devices	Waste spend (€/yr)	Realisable value (€/yr)	Per device per month	Share of spend
16+	8	238	2,705,068	225,055	€79	8.3 %
6–15	12	115	1,384,780	74,166	€54	5.4 %
2–5	32	94	1,096,332	72,285	€64	6.6 %

Computed per customer (each customer's realisable value divided by its devices), the value per device per month has a median of €38 and quartiles of €25 and €93 across the 52 customers; weighted by the 447 devices instead, the upper quartile is about €86. Against that, the weighing hardware from our concept costs, spread over five years:

- **Factory-fitted** (10 t cells, module, junction box, RFID reader and tags): ≈ €1,300–2,950, i.e. €22–49 per device per month.
- **Retrofit** (plus frame, mounts and installation): ≈ €2,450–5,200, i.e. €41–87 per device per month.

A retrofit plus a €30 subscription needs roughly €71–117 of value per device per month, which only customers in the top quartile of per-device value (€86–93 and above) reach, and only at the lower end of the cost range. Hardware costs are list-price estimates from the weighing concept.

Commercial model

We propose a SMART software tier plus a weighing hardware option, with an outcome-based start for pilots; all parameters below are our hypotheses, run through the official calculator (`baselines/business_case.py`, files in `solution/business_case/`).

Offer	Who	Charging	Why
Service Audit (software)	SMART customers with weighing, first those with 16+ devices	€15 / 30 / 45 per device per month (scenarios)	value is per device and recurs; fits SMART's existing subscription
Weighing, factory-fitted	all new compactors	in the machine price	marginal cost €22–49 per device per month over five years
Weighing retrofit	top-quartile devices by value	one-off kit + installation, or a lease	€41–87 per device per month only pays where value is high

Offer	Who	Charging	Why
Outcome-based pilot	first customers, 12 months	25 % of savings verified in the pilot (modelled here on realisable value)	customer pays only from proven results; builds the evidence base

Cost to serve and adoption (hypotheses):

- Onboarding €800 per customer (contract and invoice mapping) plus €50 per device.
- Operating €4 per device per month (cloud, data) and €40 per customer per month support.
- A customer buys only if realisable value is at least 3× the annual fee; 10 %, 20 % or 35 % of eligible customers adopt.

Europress stays neutral: it sells evidence and decision support to the customer and does not take a position on operators' prices.

Sensitivity

A higher price earns more per customer but shrinks the eligible group fast: 22 customers pass the 3× value test at €15, 14 at €30 and 7 at €45. Year-1 contribution is negative at €15 (onboarding is not covered) and positive at €30–45. The synthetic portfolio has only 52 customers, so read the unit economics, not the totals.

Price (€/device/month)	Eligible customers	Adoption	Devices	Revenue (€/yr)	Operating cost (€/yr)	Onboarding (€)	Year-1 contribution (€)
15	22	10 %	24	4,392	2,227	2,980	−815
15	22	35 %	85	15,372	7,795	10,430	−2,853
30	14	10 %	10	3,564	1,147	1,615	802
30	14	35 %	35	12,474	4,015	5,652	2,806
45	7	10 %	6	3,240	624	860	1,756
45	7	35 %	21	11,340	2,184	3,010	6,146
25 % of realisable value (modelled)	52	10 %	45	9,288	4,642	6,395	−1,749
25 % of realisable value (modelled)	52	35 %	156	32,507	16,246	22,382	−6,121

The outcome-based option keeps every customer eligible (they always keep 75 % of the value) and turns positive from year 2, when onboarding is paid: €4,646 to €16,261 a year at 10–35 % adoption. The calculator does not compute revenue for a savings share (its outcome CSV shows annual_revenue_eur = 0, which is not a forecast), so those two rows are our own calculation from its per-customer output. No saving in them is certified by a customer: they are synthetic realisable values times our assumed shares. The 20 % adoption rows are in `business_case/main/business_case_scenarios.csv`.

Productisation in SMART and future models

The service reuses what SMART already has (connectivity, device records, the customer portal) and adds three pieces: mass in the telemetry, a document intake for contracts and invoices, and an action inbox where a person approves each proposal. The working demo (`solution/app.py`) shows that flow end to end on synthetic data.

1. **Now (pilot):** weighing on 20–30 units, audit engine run monthly, outcome-based pricing, results reviewed with each customer.
2. **Next (product):** Service Audit tier in the SMART portal; e-invoice intake to remove manual uploads; scorecard and €/t reports; ESG tonnage export.
3. **Later (new roles, hypotheses):** models where Europress sits in the middle of the value chain.

With the future models switched on in the calculator (operator tendering as a service, transport procurement at a 5 % logistics discount with 30 % realisation, a tonne-based full service at a 2 % margin, a materials marketplace adding €5/t at 30 % realisation), more customers become eligible at every price: 31 instead of 22 at €15, and 17 instead of 14 at €30. These roles differ from Europress's position today and are labelled as hypotheses.

Risks, assumptions and components

The biggest commercial risk is the operator relationship: the service must read as transparency, not as a weapon, because operators may also be Europress's channel partners.

- **Neutrality:** findings are called discrepancies, not accusations; part of every difference comes from sensors, combined loads or data gaps. Operators also gain: fewer disputes, verified tonnage, and a scorecard that shows good performance.
- **Human approval:** nothing is sent to an operator automatically; every proposal waits for the customer's decision and is logged.
- **Onboarding friction:** Europress does not know contract prices, so customers must provide contracts and invoices; e-invoice intake is the fix.

What remains unproven.

- Willingness to pay at any price: no customer research was done.
- Realisation shares, adoption and the 3× purchase rule are our hypotheses.
- That operators share weighbridge data, accept renegotiation, or that tenders succeed.
- That the weighing concept reaches σ ⩽ 1.5–2 % on real machines (see the weighing concept's validation plan).
- Future models assume roles Europress does not have today.

Third-party components (all used unmodified as libraries):

Component	Use	Licence
Since AI synthetic dataset and starter code (v1.0.1)	data, baselines, scoring harness, business-case calculator	Since AI Hackathon 2026 rules
pandas, NumPy, scikit-learn	data processing, baseline ML for Task B	BSD-3-Clause
PyArrow	Parquet reading	Apache-2.0
DuckDB, PyYAML, pytz, tabulate	queries, configuration, time zones, tables	MIT
Streamlit	demo dashboard	Apache-2.0
Vega-Altair	dashboard charts	BSD-3-Clause

No external datasets, pretrained models or paid services are used. All of these licences permit commercial use.